KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6492C **6 Credits: 4** **Program Elective** **Source-grounded**

Entertainment Electronics

● The course aims to provide broad knowledge on various industry standards, algorithms

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6492C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6492C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electronics and Computer Engineering
Course code	6492C
Course title	Entertainment Electronics
Semester	6 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

7 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● The course aims to provide broad knowledge on various industry standards, algorithms
- and technologies used to carry out digital audio and video broadcasting in the
- infotainment industry.

Course outcomes

CO	Official outcome	Study level
CO1	Understand packetized streaming of digital media 14 Understanding happens in the field of infotainment industry.	See official syllabus
CO2	Realise the critical aspects of DVB standards used 17 Understanding for media broadcasting in infotainment industry.	See official syllabus
CO3	Realise the critical aspects of DAB standards used 13 Understanding for media broadcasting in infotainment industry. Understand modern display technologies for video 14	See official syllabus
CO4	Understanding reproduction.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3
CO2	CO2 3 3 2
CO3	CO3 3 3 2
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	infotainment industry.	2	As specified
Module 2	Realise the critical aspects of DVB standards used for media broadcasting	2	As specified
Module 3	Realise the critical aspects of DAB standards used for media broadcasting	2	As specified
Module 4	Understand modern display technologies for video reproduction.	1	As specified

MODULE 1

infotainment industry.

This module is presented from the official syllabus scope for Entertainment Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: infotainment industry.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ 6 Explain Digital media streaming: Understanding

6 Explain Digital media streaming: Understanding is an official topic in Module 1 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Explain Digital media streaming: Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 explain digital media streaming: understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Explain Digital media streaming: Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Explain Digital media streaming: Understanding □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes.

8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes. is an official topic in Module 1 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 ● brief review of analog television: scanning, horizontal and vertical synchronization, color information, transmission methods.ntsc and pal standards. ● digital media streaming: packetized elementary stream of audio-video data, mpeg data stream, mpeg-2 transport stream packet, accessing a program, scrambled programs, program synchronization.psi, additional (network information and service description)information in data streams for set-top boxes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Realise the critical aspects of DVB standards used for media broadcasting

This module is presented from the official syllabus scope for Entertainment Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Realise the critical aspects of DVB standards used for media broadcasting
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▾ Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding

Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding is an official topic in Module 2 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain cable tv broadcasting 4 understanding explain terrestrial tv broadcasting understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▾ Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system.

Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system. is an official topic in Module 2 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate broadcasting for handheld devices 4 understanding ● satellite tv broadcasting –dvb-s parameters, dvb-s modulator, dvb-s set-top box, dvb-s2. ● cable tv broadcasting – dvb-c standard, dvb-c modulator, dvb-c set-top box. ● terrestrial tv broadcasting – dvb-t standard, dvb-t modulator, dvb-t carriers and system parameters, dvb-t receiver. ● broadcasting for handheld devices – dvb-h standard dvb tele-text, dvb subtitling system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Realise the critical aspects of DAB standards used for media broadcasting

This module is presented from the official syllabus scope for Entertainment Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Realise the critical aspects of DAB standards used for media broadcasting
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▾ Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5

Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5 is an official topic in Module 3 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain digital audio broadcasting (dab) 8 understanding outline digital radio mondiale (drm) 5 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates.

Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates. is an official topic in Module 3 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding ● digital audio broadcasting (dab): comparison of dab with dvb. physical layer of dab. dab modulator, dab data structure, dab single frequency networks, data broadcasting using dab. ● digital radio mondiale (drm): transmitter and receiver, data rates. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Understand modern display technologies for video reproduction.

This module is presented from the official syllabus scope for Entertainment Electronics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand modern display technologies for video reproduction.
- Official duration: As specified
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

▾ Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/

Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3

https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ is an official topic in Module 4 of Entertainment Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline television of future 6 understanding ● display technology: block diagram of video reproduction system in a tv, cathode ray tubes, basic principle of plasma displays, lcd displays, light-emitting diode displays, field emission displays, organic light emitting device displays. ● television of future: holographic tv, virtual reality, augmented reality. text / reference t/r book title/author w. fischer, digital video and audio broadcasting technology: a practical t1 engineering guide (signals and communication technology), springer, 2020 lars-ingemar lundström, understanding digital television an introduction to t2 dvb systems with satellite, cable, broadband and terrestrial tv, focal press, elsevier,2006. k f ibrahim, newnes guide to television and video technology, newnes, t3 2007. t/r book title/author t4 jiun-haw lee, david n. liu, shin-tson wu, introduction to flat panel displays, wiley, 2008. r1 wolfgang hoeg, thomas lauterbach, digital audio broadcasting: principles and applications of dab, dab+ and dmb, wiley, 2009. r2 john watkinson, introduction to digital audio, focal press, 1994. r3 john watkinson, introduction to digital video, focal press, 2001. online resources sl.no website link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/iot/homepage 5 http://esp32.net/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps → Application →
Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure →
Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 6 Explain Digital media streaming: Understanding. (3 marks)**

▶ **Define or explain 8 ● Brief Review of Analog Television: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods.NTSC and PAL standards. ● Digital media streaming: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization.PSI, Additional (Network information and service description)information in data streams for set-top boxes.. (3 marks)**

Module 2

▶ **Define or explain Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding. (3 marks)**

▶ **Define or explain Illustrate broadcasting for Handheld devices 4 Understanding ● Satellite TV broadcasting –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2. ● Cable TV broadcasting – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. ● Terrestrial TV broadcasting – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. ● Broadcasting for Handheld devices – DVB-H Standard DVB tele-text, DVB subtitling system.. (3 marks)**

Module 3

▶ **Define or explain Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5. (3 marks)**

▶ Define or explain Understanding ● Digital Audio Broadcasting (DAB): Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. ● Digital Radio Mondiale (DRM): Transmitter and receiver, Data rates.. (3 marks)

Module 4

▶ Define or explain Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer, Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **6 Explain Digital media streaming: Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain cable TV broadcasting 4 Understanding Explain terrestrial TV broadcasting Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain Digital Audio Broadcasting (DAB) 8 Understanding Outline Digital Radio Mondiale (DRM) 5**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Outline television of future 6 Understanding ● Display Technology: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. ● Television of future: Holographic TV, Virtual Reality, Augmented Reality. Text / Reference T/R Book Title/Author W. Fischer,**

Digital Video and Audio Broadcasting Technology: A Practical T1 Engineering Guide (Signals and Communication Technology), Springer, 2020 Lars-Ingemar Lundström, Understanding Digital Television An Introduction to T2 DVB Systems with Satellite, Cable, Broadband and Terrestrial TV, Focal Press, Elsevier,2006. K F Ibrahim, Newnes Guide to Television and Video Technology, Newnes, T3 2007. T/R Book Title/Author T4 Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, Introduction to Flat Panel Displays, Wiley, 2008. R1 Wolfgang Hoeg, Thomas Lauterbach, Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB, Wiley, 2009. R2 John Watkinson, Introduction to Digital Audio, Focal Press, 1994. R3 John Watkinson, Introduction to Digital Video, Focal Press, 2001. Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/interne of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/106/105/106105166/
- https://www.raspberrypi.org/blog/getting-started-with-iot/
- https://learn.adafruit.com/category/internet-of-things-iot https://www.arduino.cc/en/IoT/HomePage http://esp32.net/

IN-PAGE SEARCH

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